

FoundationStereo: Zero-Shot Stereo Matching

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Motivation

- Many deep stereo matching models still rely on **per-domain fine-tuning** to achieve competitive performance.
- **Sim-to-real gap:** Synthetic training fails on real scenes
 - **Challenging structures:** Reflections, textureless, thin objects
 - **Weak zero-shot:** Poor generalization to unseen domains

Key Contributions

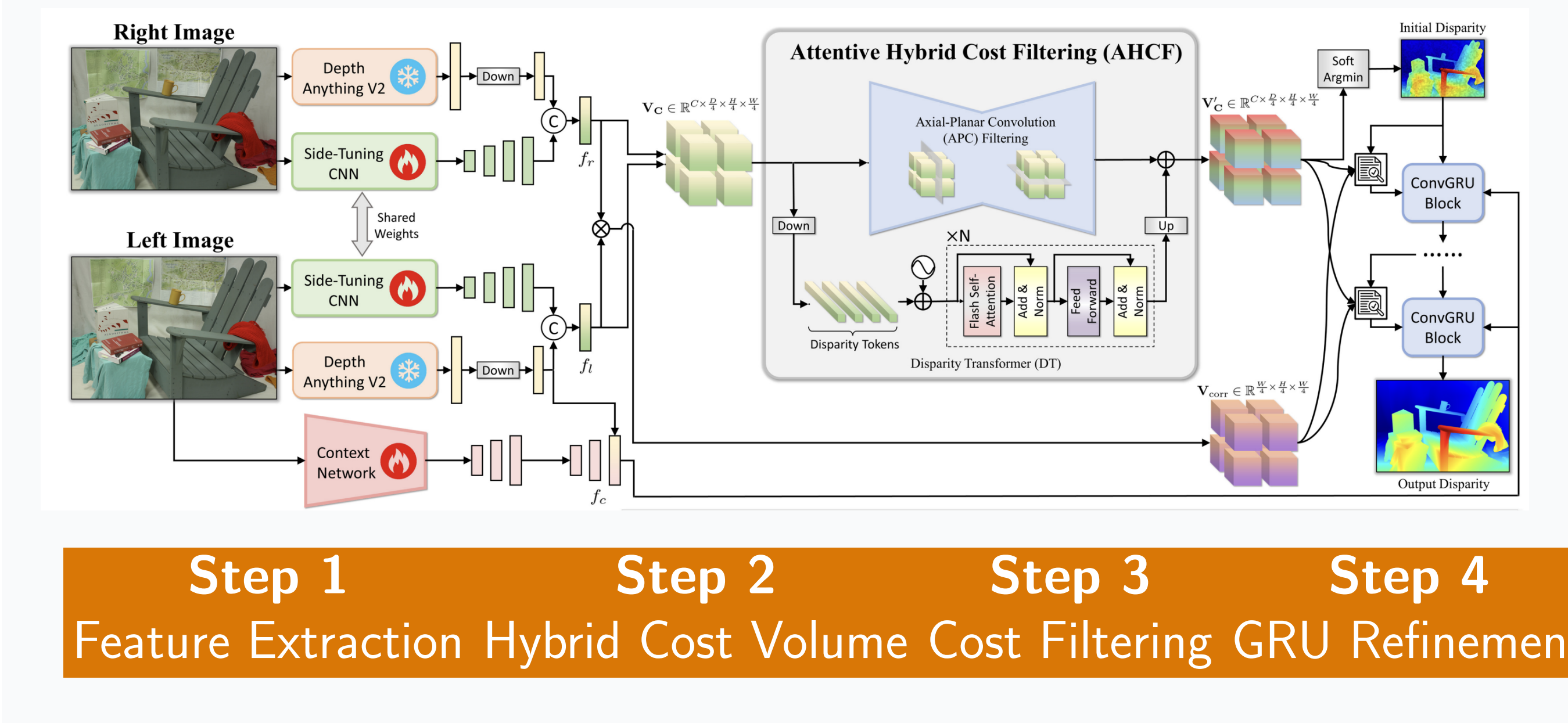
1. **FoundationStereo:** Stereo foundation model with strong zero-shot generalization
2. **FSD Dataset:** 1M synthetic stereo pairs with high diversity
3. **Side-Tuning Adapter:** Adapts DepthAnythingV2 for stereo matching
4. **Attentive Hybrid Cost Filtering:** Local + global reasoning

FSD Dataset

- 1 million stereo pairs** rendered via NVIDIA Omniverse with high realism.
- 1280×720** Resolution **1M** Stereo Pairs
- **Scenes:** Indoor, outdoor, driving, flying
 - **Diversity:** Baselines, intrinsics, lighting
 - **Self-curation:** Removes ambiguous synthetic stereo pairs

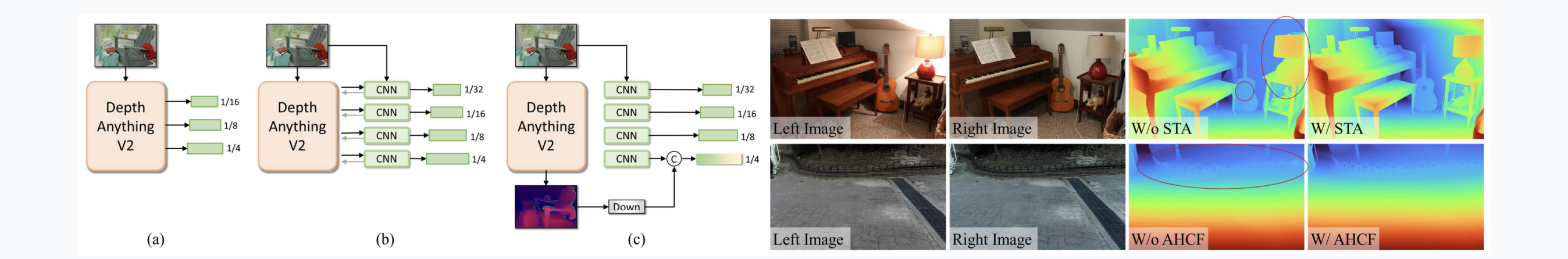


Architecture Overview



Key Ideas

- **Side-Tuning Adapter (STA)**
Adapts monocular depth features from **DepthAnythingV2** for stereo matching through lightweight adapters.
- **Attentive Hybrid Cost Filtering**
 - **Axial-Planar Conv:** Efficient spatial + disparity cost-volume filtering
 - **Disparity Transformer:** Long-range reasoning over disparity space



Zero-Shot Results

State-of-the-art zero-shot performance without target-domain fine-tuning.

Dataset	Metric ↓	FoundationStereo
Middlebury	BP-2	1.1
ETH3D	BP-1	0.5
KITTI-12	D1	2.3
KITTI-15	D1	2.8

Lower is better. Values are error rates (%).

In-the-Wild Results

Comparison of stereo matching results across different methods: Left RGB, CroCo v2, CREStereo, IGEV, Selective-IGEV, and Ours. The results show sharp, detailed disparity maps with accurate boundaries, particularly in challenging scenes like reflections, thin structures, and textureless regions.

Key Takeaways

- ✓ **Foundation model paradigm** works for stereo
- ✓ **Large-scale synthetic data** + monocular priors enable zero-shot transfer
- ✓ **Hybrid reasoning** crucial for robust cost volumes

Limitations

- **Efficiency:** Transformer modules need optimization for real-time
- **Transparent objects:** Limited glass/water diversity in training